MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

DEPARTMENT OF CIVIL ENGINEERING

III Semester B.Tech. (CIVIL ENGINEERING)

END SEM EXAM

Course: Surveying **Course Code: CIE-2122**

Time: 02:30PM **Date: 25/11/2024** **Max. Marks: 50**

Q. No.	Question	Marks	NBA		BL
			CO	PO	
1A	A tacheometer is set up at a point A on level ground and sights towards a staff held vertically at point B. The line of sight is inclined at an angle of $+10^{o}$ with the horizontal. The stadia readings taken on the staff are as follows: Upper stadia reading: 1.5 m Middle stadia reading: 1.2 m Lower stadia reading: 0.9 m Calculate horizontal distance, RL of point B and Gradient between point A and B. Take RL of line of collimation of point A is 120.5m, K=100 and C=0.	**5**	2		**3**
1B	the following consecutive readings were taken with a dumpy level, the instrument have been shifted after the second, fourth and seventh readings: 0.9, 1.25, 2.4, 1.375, 2.945, 3.125, 3.725, 0.1, 1.975, 2.025 and 1.775. The first reading was taken with the staff held on benchmark of elevation 100m. Enter the reading in the level book and calculate RL using rise and fall method. Apply the usual check.	**3**	2		**3**
1C	Differentiate between WCB and QB	**2**	1		**2**
2A	What are the requirement of sounding? And Describe briefly how the soundings are located by two angles from the shore and intersecting ranges.	**5**	3		**2**
2B	With the neat sketch describe the graphical method of plotting sounding location in hydrographic surveying.	**3**	3		**2**
2C	Illustrate and describe the transfer of surface alignment to underground passageway in underground surveying with the neat sketch.	**2**	3		**2**

3A	the ground Length of a line AB is know to be 540m and the elevation of A and B are respectively 450m and 350m above M.S.L. on a vertical photograph taken with a camera having focal length of 22 cm include the image a and b of these points, and there photographic co-ordinates are : (x_a= 2.6 cm, y_a=1.3 cm); (x_b= -1.9 cm, y_b=3.6 cm). the distance ab scaled directly from the photograph is 5cm. compute the flying height above the mean sea level.	**5**	4		**3**
3B	What are the advantages and limitations of total station in EDM application.	**3**	4		**2**
3C	With the neat sketch illustrate and describe the graphical method of finding horizontal and vertical angle between points using photographic measurements.	**2**	4		**2**
4A	An area 15 km x 30 km is to be photographed with 150 mm focal length camera. Determine the flight plan for a scale of 1:25000 effective at an elevation of 400 m. Take the end lap as 65% and side lap as 35%. The size of the photograph is 200 mm x 200 mm. An intervalometer will be used to control the interval between exposures with the least count of 0.5sec. The speed of the aircraft will be maintained at 200 km/hr.	**5**	4		**3**
4B	A line AB appears to be 12 cm on a photograph for which the focal length is 150mm. the corresponding line measures 2.9 cm on a map which is to a scale of 1:45000. The terrain has an average elevation of 250m above mean sea level. Calculate the flying altitude of the aircraft, above mean sea level, when the photograph was taken.	**3**	4		**3**
4C	What is Datum scale and average scale?	**2**	4		**2**
5A	What are the key components of GPS? Explain in detail.	**5**	5		**3**
5B	What is GIS? And what are its components?	**3**	5		**2**
5C	What are the steps involved in drone surveying process?	**2**	5		**2**